25 / 02 / 2026.

Si $\{X_n\}_{n\geq 0}$ Cadena de Markov con matriz de transición

$$P = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

→ Irreducible
→ $d(1) = 1$ "MCD"
→ Aperiódica

$S' = \{1, 2, 3\}$

→ Resultado ergódico

$$r_j = \frac{1}{\pi_j}$$

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$r_j = \mathbb{E}[T \mid X_0 = j]$ donde $T = \min\{n \geq 0 \mid X_n = j\}$

$t_j = \mathbb{E}[T \mid X_0 = j]$ donde $T = \min\{n \geq 0 \mid X_n \in A\}$ $A \subseteq S'$

$$t_K = 1 + \sum_{j \in S'} P_{jK}\, t_j \qquad K \in S', A$$

- Ecuaciones de Chapman — Kolmogorov. $i \longleftrightarrow j \longleftrightarrow K$
- Distribución estacionaria.

$\pi(\pi_1, \pi_2, \pi_3)$ $\pi_j > 0$ $\forall j$

Por definición del teorema una Raíz $= 1$

$\pi_1 + \pi_2 + \pi_3 = 1$ La fila tiene que sumar 1

$\pi P = \pi$

$$(\pi_1, \pi_2, \pi_3) \begin{pmatrix} 1/4 & 1/2 & 1/4 \\ 1/3 & 0 & 2/3 \\ 1/2 & 0 & 1/2 \end{pmatrix} = \begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{pmatrix}$$

$\frac{1}{4}\pi_1 + \frac{1}{3}\pi_2 + \frac{1}{2}\pi_3 = \pi_1 \Rightarrow \pi_3 = \frac{14}{12}\pi_1 = \frac{7}{6}\pi_1 = \frac{3}{8}$

$\frac{1}{2}\pi_1 + 0\pi_2 + 0\pi_3 = \pi_2 \Rightarrow \frac{1}{2}\pi_1 = \pi_2 \Rightarrow \frac{3}{16}$

$\frac{1}{4}\pi_1 + \frac{2}{3}\pi_2 + \frac{1}{2}\pi_3 = \pi_3 \Rightarrow \pi_3 = \frac{7}{6}\pi_1 = \frac{7}{6}\left(\frac{3}{8}\right) = \frac{7}{16}$